

YIFAN JIANG

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WORK EXPERIENCE

Apple, Seattle, USA

March, 2024 – Present

Senior Research Scientist at AI/ML

EDUCATION

The University of Texas at Austin, Austin, USA

2020 – 2024

Ph.D. in Electrical and Computer Engineering¹

- Advisor: Zhangyang (Atlas) Wang
- Committee: Alan Bovik, Sandeep Chinchali, Mingyuan Zhou, David Z. Pan

Research Interests: 3D Computer Vision, Neural Rendering, Machine Learning

Huazhong University of Science and Technology, Wuhan, China

2015 – 2019

B.E. in Electronic Information Engineering

PUBLICATIONS

(* indicates equal contribution)

- [ICLR] Y. Jiang, H. Tang, J. Chang, D. Xu, L. Song, Z. Wang and L. Cao, “Efficient-3DiM: Learning a Generalizable Single-image Novel-view Synthesizer in One Day”, *International Conference on Learning Representations*, 2024
- [CVPR] Y. Jiang, H. Peter, B. Mildenhall, D. Xu, J. T. Barron, and Z. Wang, “AligNeRF: High-Fidelity Neural Radiance Fields via Alignment-Aware Training”, *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023
- [ECCV] Y. Jiang*, D. Xu*, P. Wang, Z. Fan, H. Shi, and Z. Wang. “SinNeRF: Training Neural Radiance Fields on Complex Scenes from a Single Image”, *European Conference on Computer Vision*, 2022.
- [ECCV] Y. Jiang*, Z. Fan*, P. Wang*, X. Gong, D. Xu, and Z. Wang. “Unified Implicit Neural Stylization”, *European Conference on Computer Vision*, 2022.
- [ECCV] Y. Jiang, B. Wronski, B. Mildenhall, J. T. Barron, Z. Wang, and T. Xue. “Fast and High-Quality Image Denoising via Malleable Convolutions”, *European Conference on Computer Vision*, 2022.
- [WACV] Y. Jiang, X. Gong, J. Wu, H. Shi, Z. Yan, and Z. Wang, “AutoX3D: Searching Ultra-Efficient Architecture for Video Understanding”, *IEEE Winter Conference on Applications of Computer Vision*, 2022.
- [NeurIPS] Y. Jiang, S. Chang, and Z. Wang, “TransGAN: Two Pure Transformers can Make One Strong GAN and That Can Scale Up”, *Advances in Neural Information Processing Systems*, 2021.
- [ICCV] Y. Jiang, H. Zhang, J. Zhang, Y. Wang, Z. Lin, K. Sunkavalli, S. Chen, S. Amirghodsi, S. Kong, and Z. Wang, “SSH: A Self-supervised Framework for Image Harmonization”, *IEEE International Conference on Computer Vision*, 2021.
- [TIP] Y. Jiang, X. Gong, D. Liu, Y. Cheng, C. Fang, X. Shen, J. Yang, P. Zhou, and Z. Wang, “EnlightenGAN: Deep Light Enhancement without Paired Supervision”, *IEEE Transaction on Image Processing*. **IEEE SPS Young Author Best Paper Award.**
- [ECCV] Z. Zhu, Z. Fan, Y. Jiang, Junru Wu*, Kai Wang, Gong Zhang, Humphrey Shi, Zhangyang Wang, Shiyu Chang, “Fsgs: Real-time Few-shot View Synthesis using Gaussian Splatting”, *European Conference on Computer Vision*, 2024.
- [CVPR] V. Goel, E. Peruzzo, Y. Jiang, D. Xu, N. Sebe, T. Darrell, Z. Wang, and H. Shi, “PAIR-Diffusion: Object-Level Image Editing with Structure-and-Appearance Paired Diffusion Models”, *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
- [CPAL] Q. Wu, Y. Jiang, J. Wu, K. Wang, G. Zhang, H. Shi, Z. Wang, S. Chang, “Broad Spectrum Image Deblurring via An Adaptive Super-Network”, *Conference on Parsimony and Learning (CPAL)*, 2024
- [NeurIPS] Z. Wang, Y. Jiang, Y. Lu, Y. Shen, P. He, W. Chen, Z. Wang, M. Zhou, “In-Context Learning Unlocked for Diffusion Models”, *Advances in Neural Information Processing Systems*, 2023. **Spotlight**

¹Studied at Texas A&M University from Aug. 2019 to Aug. 2020; then transferred with my advisor to UT Austin

- [**NeruIPS**] Z. Wang, **Y. Jiang**, H. Zheng, P. Wang, P. He, Z. Wang, W. Chen, M. Zhou, “Patch Diffusion: Faster and More Data-Efficient Training of Diffusion Models”, *Advances in Neural Information Processing Systems*, 2023.
- [**TIP**] Q. Wu, **Y. Jian***, J. Wu, V. Kulikov, V. Goel, N. Orlov, H. Shi, Z. Wang, S. Chang, “Broad Spectrum Image Deblurring via An Adaptive Super-Network”, *Transactions on Image Processing*, 2023
- [**TMLR**] Q. Wu, X. Chen, **Y. Jiang**, Z. Wang, “Chasing Better Deep Image Priors Between Over-and Under-parameterization”, *Transactions on Machine Learning Research*, 2023
- [**CVPR**] D. Xu, **Y. Jiang**, P. Wang, Z. Fan, Y. Wang, and Z. Wang, “NeuralLift-360: Lifting An In-the-wild 2D Photo to A 3D Object with 360° Views”, *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023. **Highlight (2.5% of 9155 submissions)**
- [**ICLR**] Z. Fan, P. Wang, **Y. Jiang**, X. Gong, D. Xu, and Z. Wang, “NeRF-SOS: Any-View Self-supervised Object Segmentation on Complex Real-World Scenes”, *International Conference on Learning Representation*, 2023
- [**NeurIPS**] D. Xu*, P. Wang*, **Y. Jiang**, Z. Fan, and Z. Wang, “Signal Processing for Implicit Neural Representations”, *Advances in Neural Information Processing Systems*, 2022.
- [**MM**] D. Xu, H. Poghosyan, S. Navasardyan, **Y. Jiang**, H. Shi, and Z. Wang, “ReCoRo: Region-Controllable Robust Light Enhancement by User-Specified Imprecise Masks”, *ACM Multimedia*, 2022
- [**TIP**] Z. Chen, **Y. Jiang**, D. Liu, and Z. Wang, “CERL: A Unified Optimization Framework for Light Enhancement with Realistic Noise”, *IEEE Transaction on Image Processing*
- [**NeurIPS**] B. Pan, R. Panda, **Y. Jiang**, Z. Wang, R. Feris, and A. Oliva, “IA-RED2: Interpretability Aware Redundancy Reduction for Vision Transformers”, *Advances in Neural Information Processing Systems*, 2021.
- [**ICLR**] T. Meng*, X. Chen*, **Y. Jiang**, and Z. Wang, “A Design Space Study for LISTA and Beyond”, *International Conference on Learning Representations*, 2021.
- [**DAC**] Y. Fu, Z. Yu, Y. Zhang, **Y. Jiang**, C. Li, Y. Liang, M. Jiang, Z. Wang, and Y. Lin, “InstantNet: Automated Generation and Deployment of Instantaneously Switchable Precision Networks”, *Design Automation Conference*, 2021.
- [**ICCV**] X. Gong, S. Chang, **Y. Jiang**, and Z. Wang. “AutoGAN: Neural Architecture Search for Generative Adversarial Networks”, *IEEE International Conference on Computer Vision*, 2019.

INTERN EXPERIENCE

Apple, Seattle, USA May. 2023 – Jan. 2024

Research Intern at AI/ML Argos Team, worked with [Liangliang Cao](#), [Hao Tang](#), and [Rick Chang](#)

- Designed an efficient diffusion model training algorithm.

Adobe, San Jose, USA May. 2022 – Dec. 2022

Research Intern at [Marc Levoy's Team](#), worked with [Zhihao Xia](#), [Cecilia Zhang](#), [Xiuming Zhang](#) [Jiawen Chen](#).

- Developed a high-quality depth estimator using multi-sensor assistance.

Google Research, Mountain View, USA May. 2021 – May. 2022

Research Intern at [GCam](#), worked with [Tianfan Xue](#), [Bart Wroński](#), [Ben Mildenhall](#), [Peter Hedman](#), [Jon Barron](#).

- Developed a fast denoising operator as well as an efficient backbone.
- Designed a high-fidelity neural radiance field that can render high-quality novel view images.

Adobe, San Jose, USA May. 2020 – Nov. 2020

Research Intern at Applied Research Team (ART), worked with [He Zhang](#) and [Jianming Zhang](#).

- Developed a self-supervised method for image harmonization.

Bytedance AI Lab, Beijing, China Jan. 2019 – Aug. 2019

Research Intern at Bytedance AI Lab, worked with [Xiaohui Shen](#), [Ding Liu](#), [Chen Fang](#), [Jianchao Yang](#).

- Designed a jointly image denoising and low-light enhancement algorithm for a selfie camera app [FaceU](#).

MEDIA HIGHLIGHT

- TransGAN was covered by [Quanta Magazine](#) (Mar. 2022) and was highlighted by [Top AI influencers](#) and high-profile [YouTubers](#), as well as considered as [the most influential new paper of the month](#) (Feb. 2021).

- AutoGAN was covered by [Synced AI Technology & Industry Review](#) (Aug. 2019), and also featured on [Towards Data Science](#) (Sep. 2019) and [Analytics Magazine](#) (Aug. 2019), etc.

COMMUNITY SERVICES

- Reviewer for: CVPR, ICCV, ECCV, ICML, NeurIPS, ICLR, Siggraph, Siggraph Aisa, IROS, WACV, IJCAI, TPAMI, TIP, IJCV, NeuroComputing, RA-L, TCI, TCSVT
- Workshop Organizer for: [ECCV RLQ-TOD Workshop 2020](#)

AWARDS

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|---|------|
| • IEEE SPS Young Author Best Paper Award | 2024 |
| • University of Texas Graduate Dean's Prestigious Fellowship | 2023 |
| • Apple Scholars in AI/ML Fellowship | 2023 |